

College of Industrial Technology

King Mongkut's University of Technology North Bangkok

Final Examination of Semester 2

Year: 2019

Subject: 392132 Physics II

Section: 15 – 18

Date: 3 March 2020

Time: 08.00-10.00

Seat No.

Name: _____ ID: _____ Class: _____

Directions: The test is designed to measure your comprehension. There will be 6 pages (including this page), 13 question and they are worth 40 points.

- Answer the questions on this test paper.
- Books, documents and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators can not be used in this test.

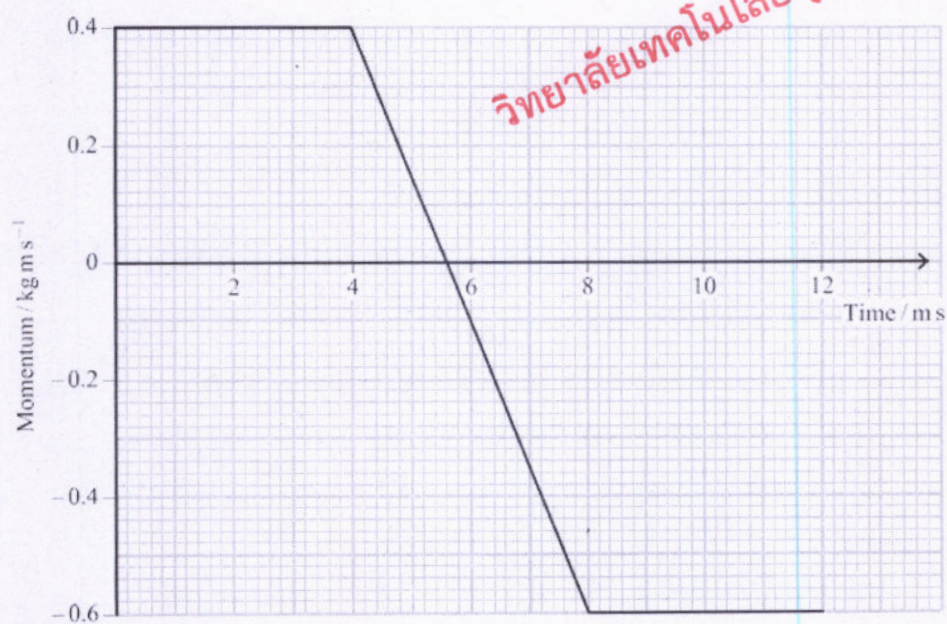
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Now begin the test.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

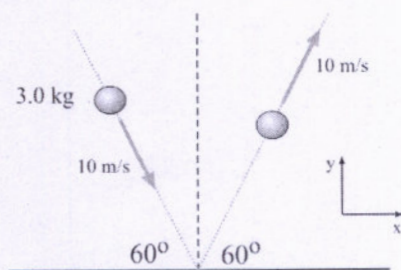
1	2	Total

1. A 120 kilogram giraffe is running at a speed of 30 kilometers per hour. Calculate the momentum of a giraffe.(2 points)
2. Show that the kinetic energy K.E. of a particle of mass m is related to the magnitude of the momentum p of that particle by $K.E. = p^2/2m$. Note: This expression is invalid for particles traveling at speeds near that of light. (2 points)
3. A tennis ball is travelling horizontally with a momentum of 0.4 kg m s^{-1} just before it is hit with a tennis racket. It rebounds horizontally from the tennis racket with a momentum of -0.6 kg m s^{-1} . The graph shows the variation in the momentum of the ball during this process.



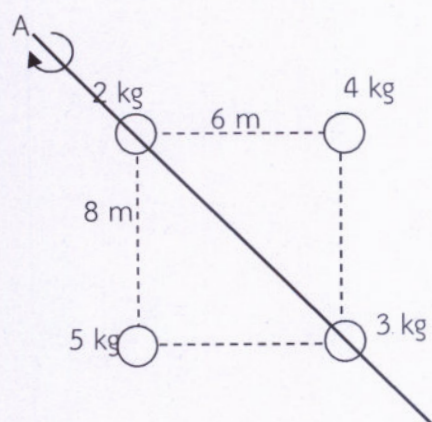
What is force exerted by the tennis ball on the racket? (3 points)

4. A 3.0 kg fish is swimming at 1.5 m/s to the right. It swallows a 0.25 kg fish swimming to the left at 4.0 m/s. What is the speed and direction of the larger fish immediately after eating? (3 points)
5. Two adjacent students stand on stationary skateboards, facing each other, and push apart. The skateboarder on the left weighs 65 kg, the one on the right is 85 kg. If the lighter skateboarder moves at 3.4 m/s, determine the speed and direction of the heavier skateboarder. (4 points)
6. A 3.0 kg steel ball strikes a wall with a speed of 10 m/s at an angle of 60° with the surface. It bounces off with the same speed and angle, as shown in Figure below. If the ball is in contact with the wall for 0.20 s, what is the average force exerted on the wall by the ball? (4 points)



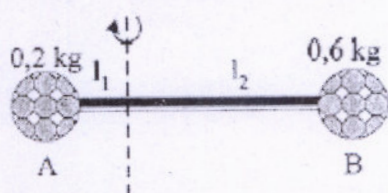
7. A 20.0-g bullet is fired horizontally into a 100-g wooden block that is initially at rest on a frictionless horizontal surface and connected to a spring having spring constant 1200 N/m. The bullet becomes embedded in the block. If the bullet-block system compresses the spring by a maximum of 50.0 cm, what was the speed of the bullet at impact with the block? (3 points)

- 8.1 From figure, find moment of inertia of this four mass rotate with A-axis. (3 points)

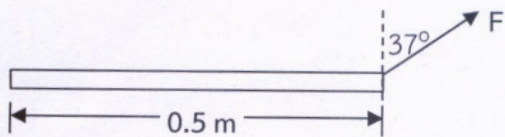


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- 8.2 Length of wire = 12 m, $l_1 = 4$ m. Ignore wire's mass. What is the moment of inertia of the system? (2 points)



9. Calculate the torque produced by the 75 N force shown in figure. (2 points)

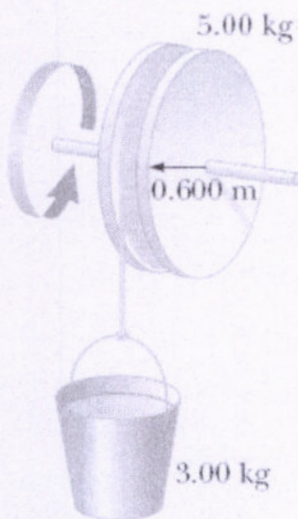


10. If the torque required to loosen a nut that holds a wheel on a car has a magnitude of 70 N.m, what force must be exerted at the end of a 35 cm lug wrench to loosen the nut when the angle of the wrench is 30° ? (3 points)

11. The motor in electric saw brings the circular blade from rest up to the rate angular velocity of 60 rev/s in $240\pi^2 \text{ rev}$. One type of blade has a moment of inertia of $1.5 \times 10^{-3} \text{ kgm}^2$. What net torque (assumed constant) must the motor apply to the blade? (3 points)

12. A blob of clay of mass M is dropped on top of a rotating disk's edge of mass $2M$ spinning at a speed of ω_0 . What will the resulting rotational speed ω be? ($I_{\text{disk}} = 0.5MR^2$) (3 points)

13. Determine the angular speed of the spool shown in Figure after the 3.00-kg bucket has fallen 5.5 m, starting from rest. The light string attached to the bucket is wrapped around the spool and does not slip as it unwinds. (3 points)



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